



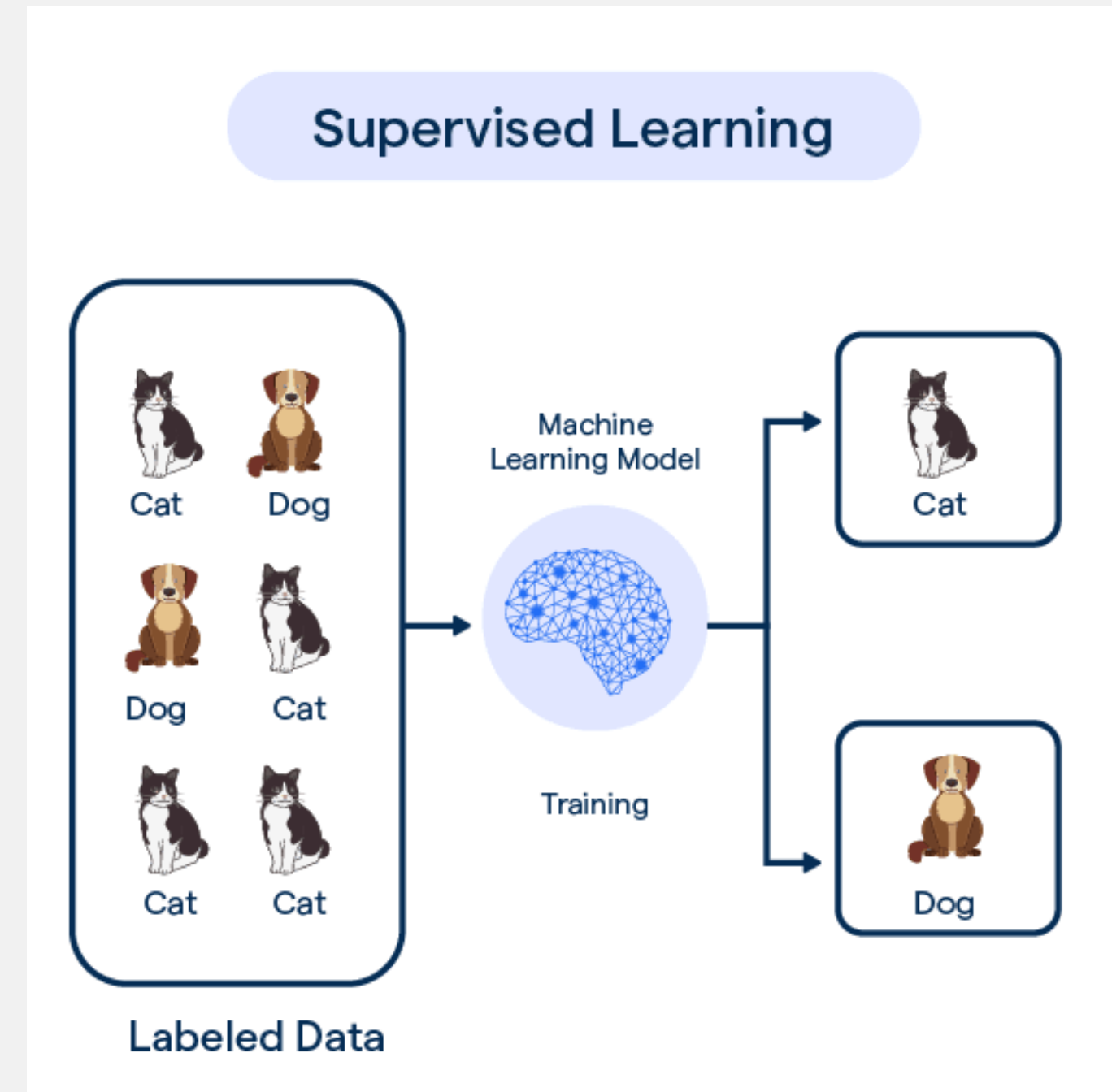
Supervised Learning pt.2

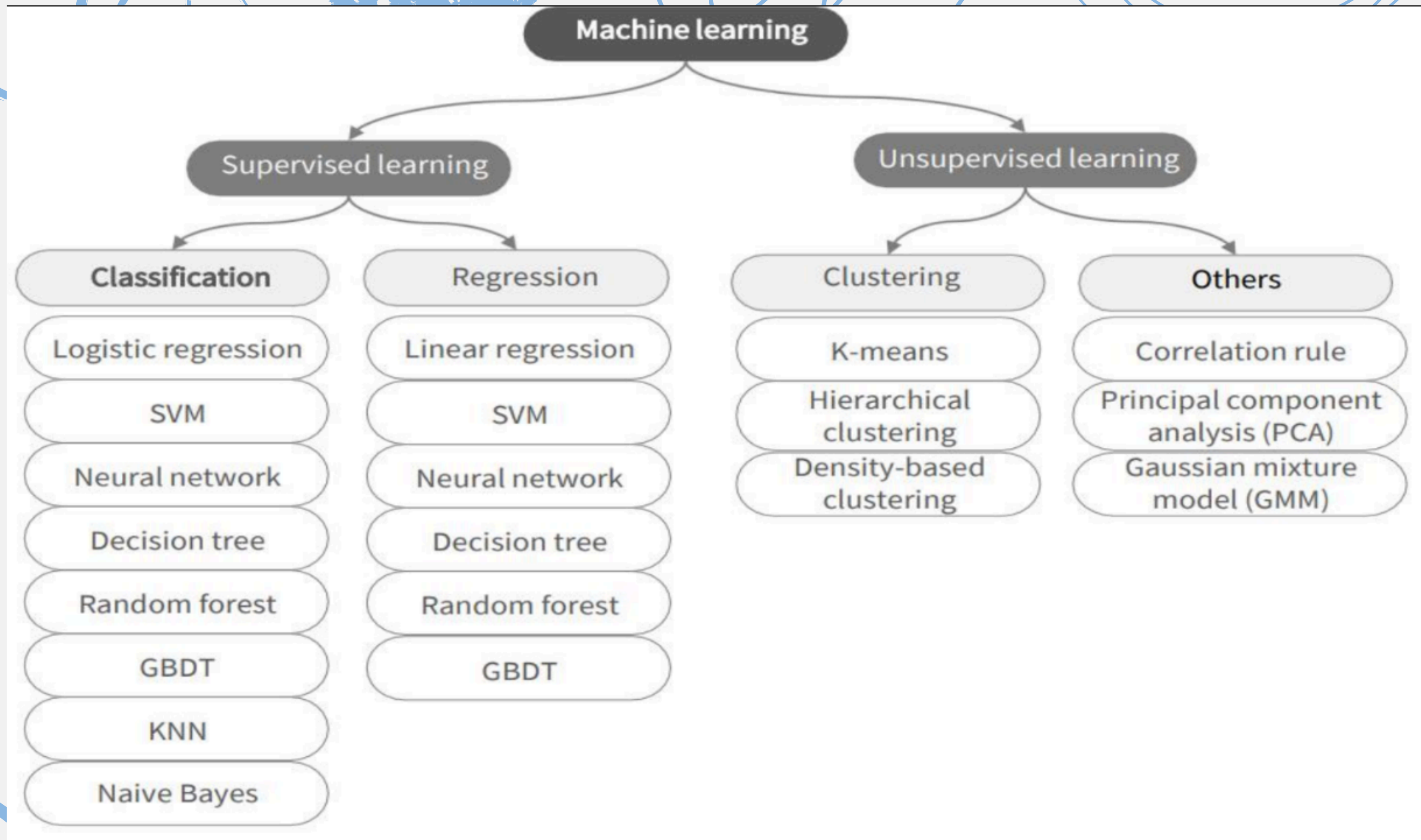
MSP'26 ML Module Session03



Supervised Learning

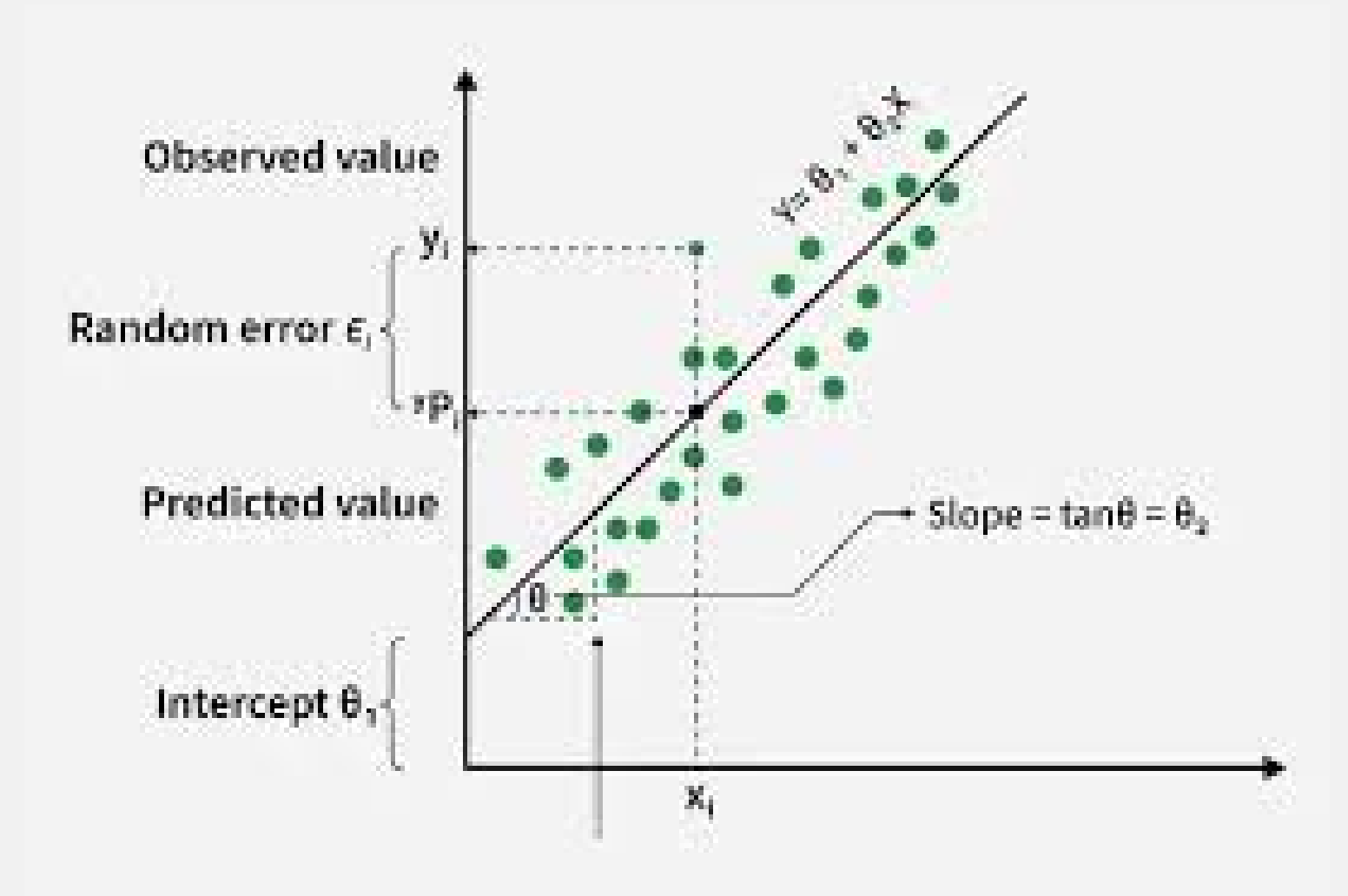
Supervised learning is a type of machine learning where a model learns from labelled data—meaning every input has a corresponding correct output. The model makes predictions and compares them with the true outputs, adjusting itself to reduce errors and improve accuracy over time. The goal is to make accurate predictions on new, unseen data.

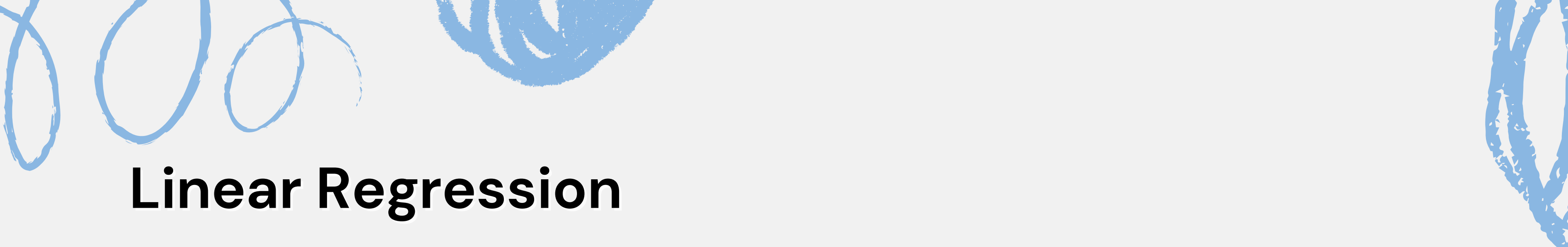




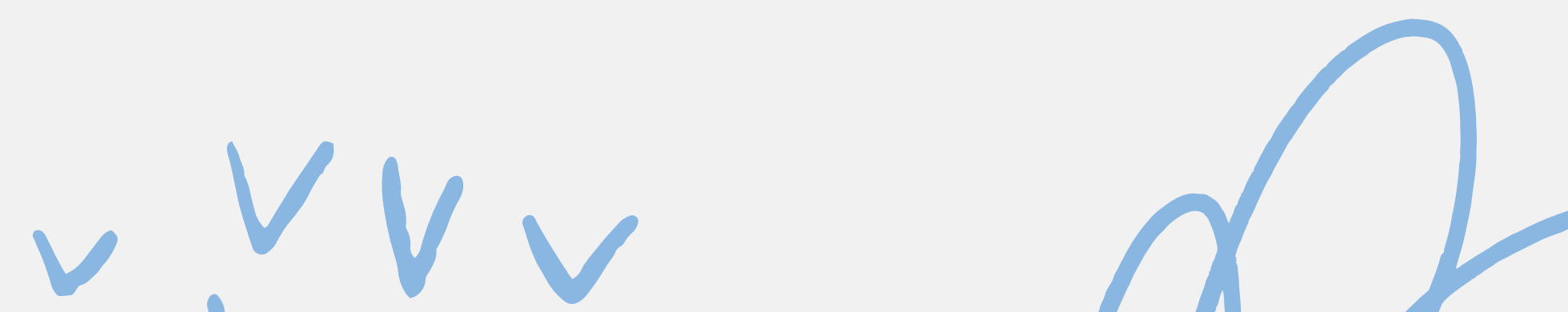
Linear Regression

Linear Regression is used to model the relationship between a dependent variable and one or more independent variables.





Linear Regression

1. Initialize parameters for slope m and intercept b
 2. Calculate the error using a cost/loss function such as MSE
 3. Find the gradient of the cost function with respect to m and b
 4. Update parameters to reduce cost/loss
 5. Repeat until the cost function converges (no longer changes or changes are very small)
- 

Linear Regression

- Dataset: (1,2), (2,4), (3,6) $n=3$
- Learning rate $\alpha=0.1$
- Initial: $m=0$, $b=0$

i	x_i	y_i	$\hat{y}_i = mx+b$	Error ($y_i-\hat{y}_i$)	$(y_i-\hat{y}_i)^2$	$x_i(y_i-\hat{y}_i)$
1	1	2	0	2	4	2
2	2	4	0	4	16	8
3	3	6	0	6	36	18
Σ	6	12	—	12	56	28

Linear Regression

$$J = \frac{1}{n} \sum (y - \hat{y})^2$$

$$J = \frac{56}{3} = 18.67$$

$$\frac{\partial J}{\partial m} = -\frac{2}{n} \sum x_i (y_i - \hat{y}_i)$$

$$= -\frac{2}{3}(28)$$

$$= -18.67$$

$$\frac{\partial J}{\partial b} = -\frac{2}{n} \sum (y_i - \hat{y}_i)$$

$$= -\frac{2}{3}(12)$$

$$= -8$$

$$m = 0 - 0.1(-18.67) = 1.867$$

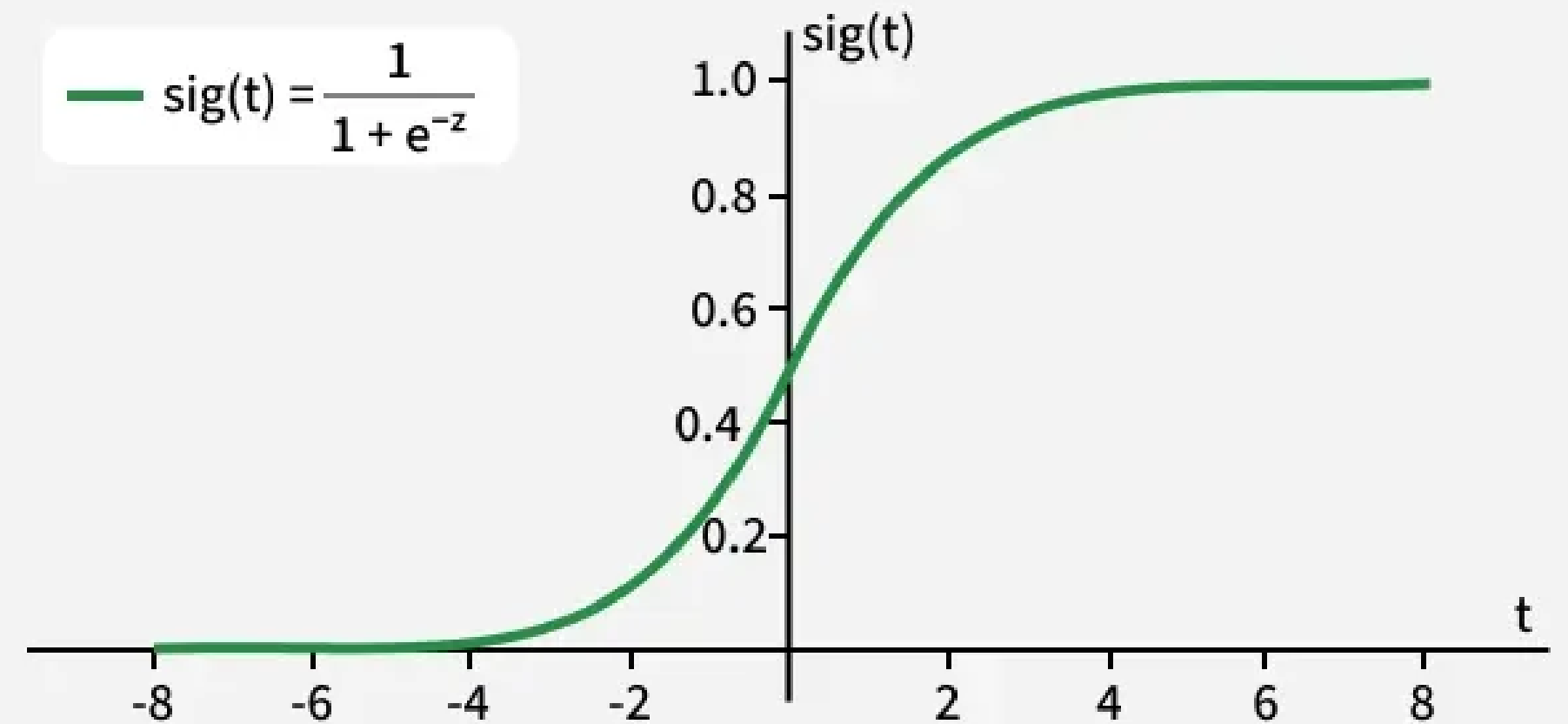
$$b = 0 - 0.1(-8) = 0.8$$

Logistic Regression

Unlike linear regression, which predicts continuous values it predicts the probability that an input belongs to a specific class.

It uses sigmoid function to convert inputs into a probability value between 0 and 1.

For more than two categories SoftMax functions are used.



Support Vector Machine (SVM)

SVM tries to find the best boundary known as hyperplane that separates different classes in the data.

The main goal of SVM is to maximize the margin between the two classes. The larger the margin the better the model performs on new and unseen data.

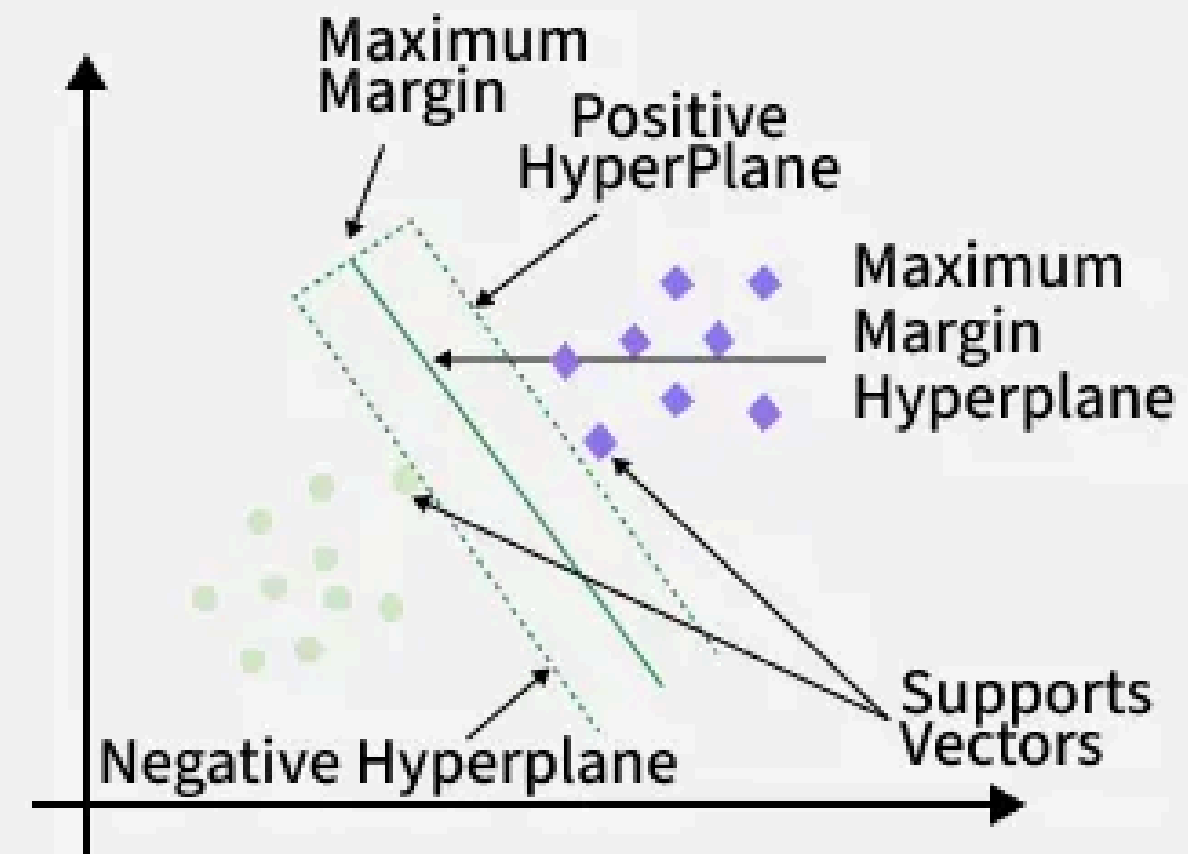
SVM is robust to outliers.

A soft margin allows for some misclassifications or violations of the margin to improve generalization. The SVM optimizes the following equation to balance margin maximization and penalty minimization:

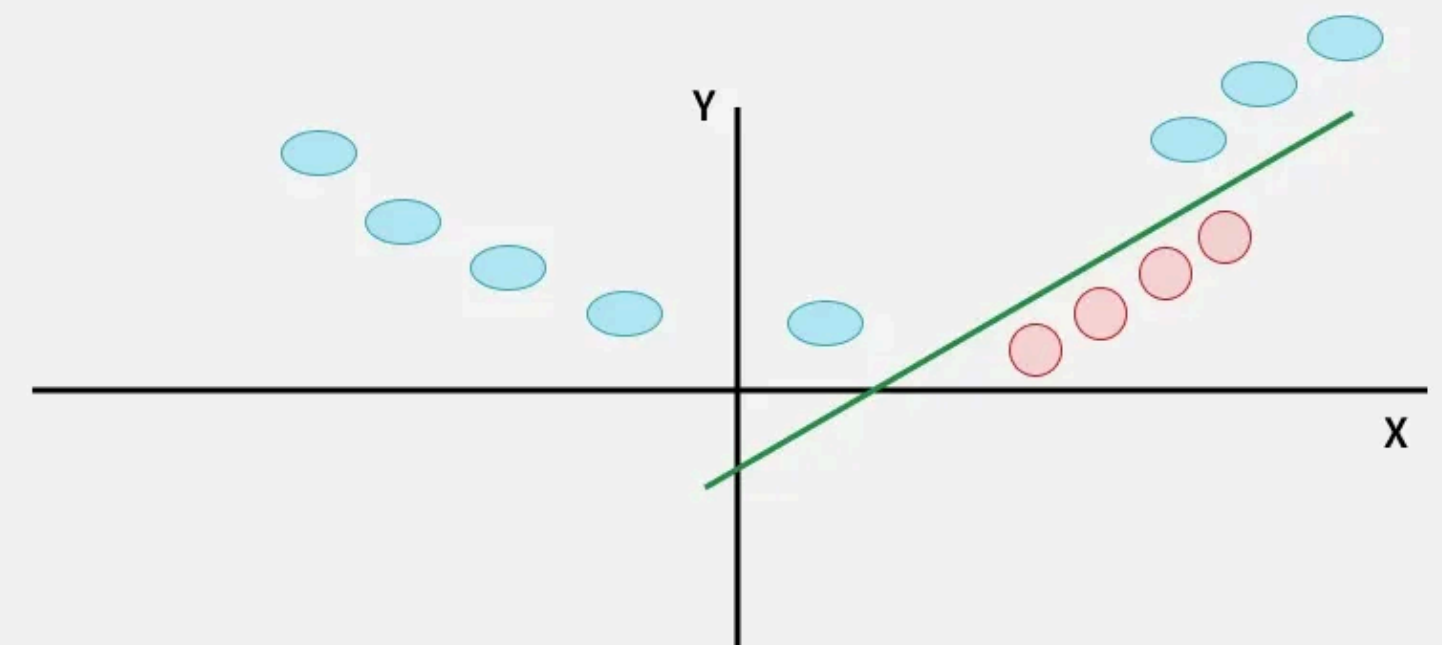
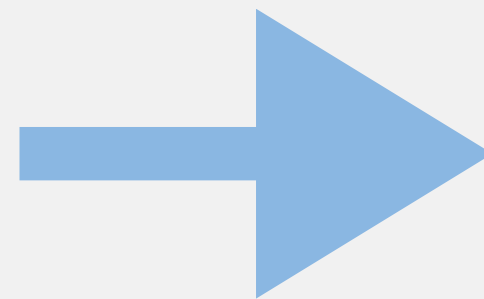
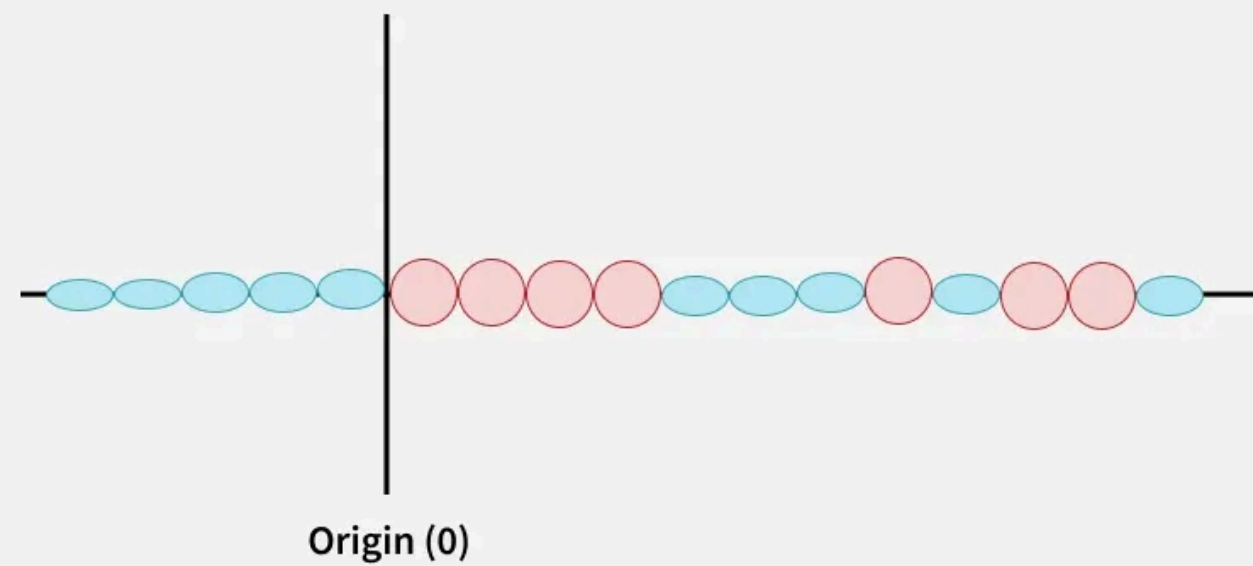
$$\text{Objective Function} = \left(\frac{1}{\text{margin}}\right) + \lambda \sum \text{penalty}$$

Support Vectors & Hyperplane

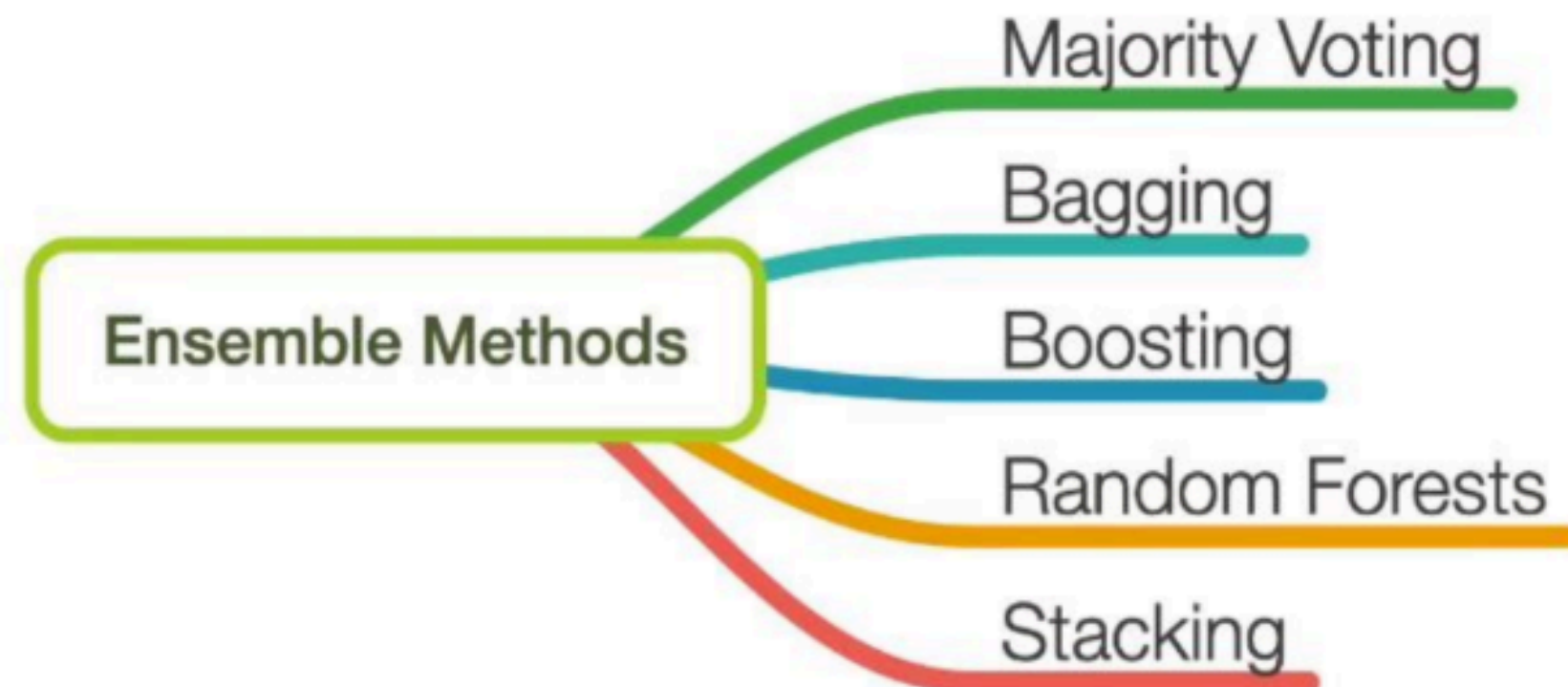
- Support Vectors are the closest data points to the hyperplane that define the class boundary.
- A hyperplane is a plane that separates different classes.



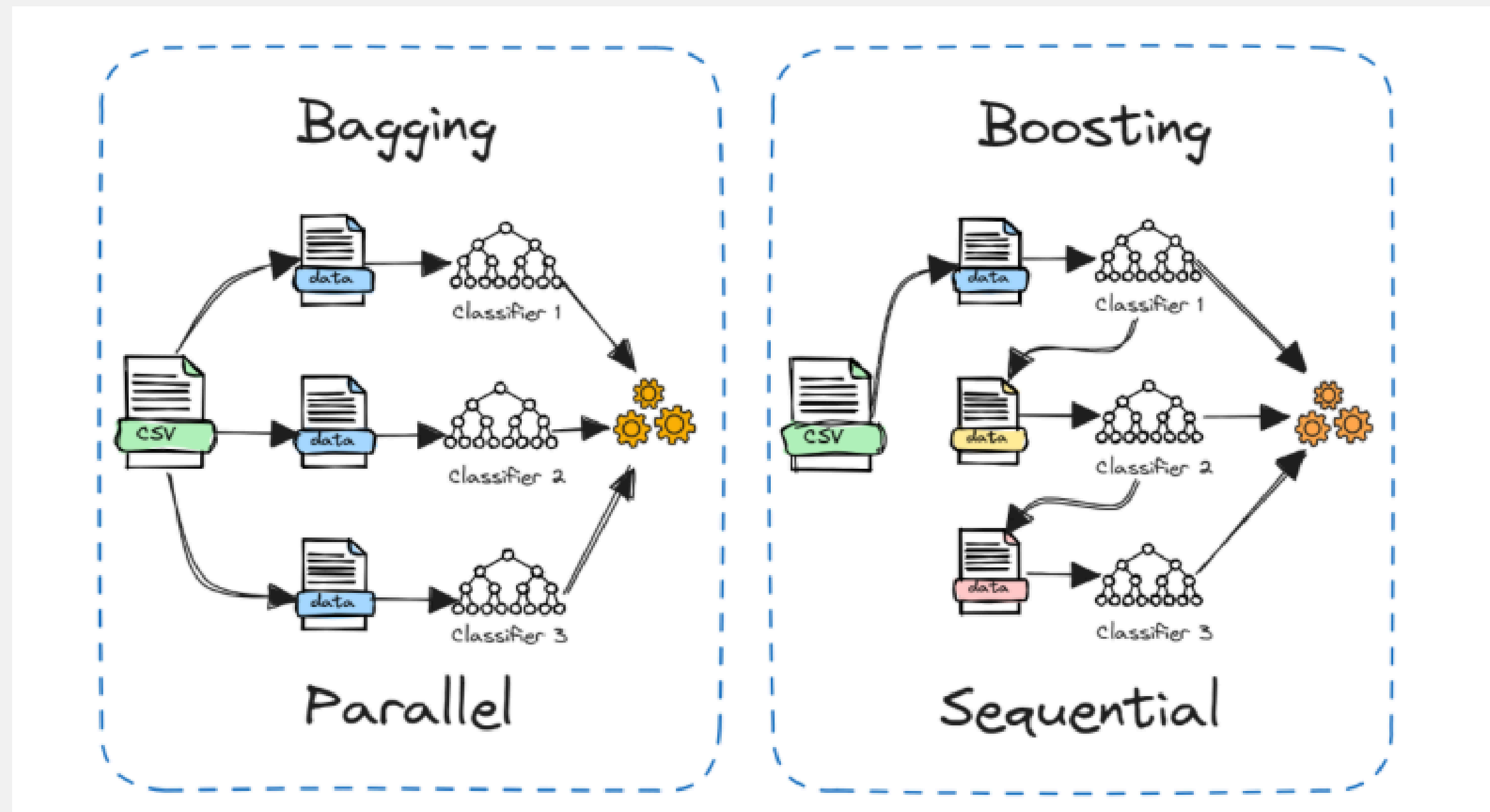
Non-linearly seperable data



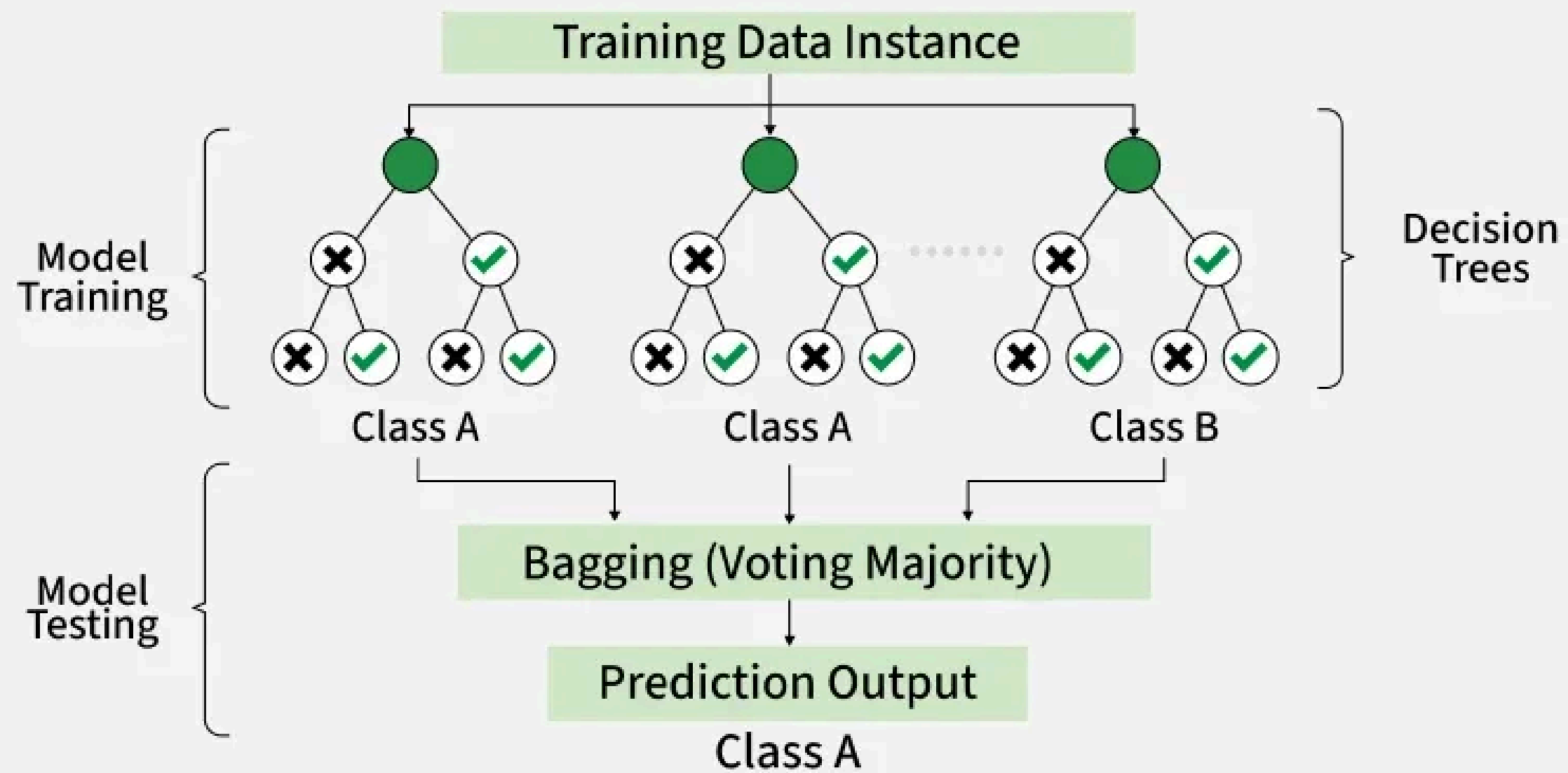
Ensemble Learning



Bagging vs Boosting



Random Forest



Stacking

also known as stacked generalization, multiple machine learning models are combined using another machine learning algorithm (combiner).

the basic idea is to train machine learning algorithms with training dataset then generate a new dataset with these models then this new dataset is used as an input for the combiner machine learning algorithm.

